BatchGuard

Capacity-aware medicine risk ranking and live clone-scan early warning

IBM Z Datathon 2026 | Theme 1: Real-Time AI for Critical Decisions | Abstract submission | Team: [Team name]

One-line summary. BatchGuard tells a drug inspector which batch to sample first this week, and why. It fuses a risk ranking learned from real CDSCO failure history with a live stream of verification-scan anomalies, and returns an explained, ranked queue that fits the inspector's limited testing capacity. A human always makes the final call.

The problem Failed and counterfeit medicines are usually found after harm has occurred. Regulators cannot test every batch, and a copied verification code can make a fake pack look genuine.	**The solution** Two evidence layers, one decision: a risk prior from real regulatory history and a live clone-suspicion detector, fused into a capacity-aware inspection queue with reason codes.
What makes it different It optimises the scarce resource, inspector sampling capacity. Success is measured as failures caught per 100 samples against simple baselines, not as a dashboard.	**What is real and what is simulated** Layer A uses real CDSCO NSQ and spurious-drug lists. Layer B is a simulated, labelled scan stream anchored to real manufacturers and batches. Both are labelled throughout.

1. The problem

- **Coldrif, September to October 2025.** At least 22 children (some reports say 24) died in Madhya Pradesh and Rajasthan after taking a cough syrup reported to contain 48.6 percent diethylene glycol against a 0.1 percent limit.
- **Slow detection.** Deaths began in September and CDSCO reported the contamination to WHO on 8 October. A central team took 19 samples from the children's doctors and nearby shops; only 4 were declared Not of Standard Quality (NSQ).
- **Nobody can test everything.** India has over 10,000 registered drug manufacturers, and enforcement falls largely on state drug controllers with limited capacity. The real question is what to test first.
- **The data exists, and so do counterfeits.** CDSCO publishes NSQ and spurious-drug lists every month (trade press reported 157 NSQ and 2 spurious samples for May 2026). A spurious drug is made by an unauthorised entity and sold under a legitimate brand.

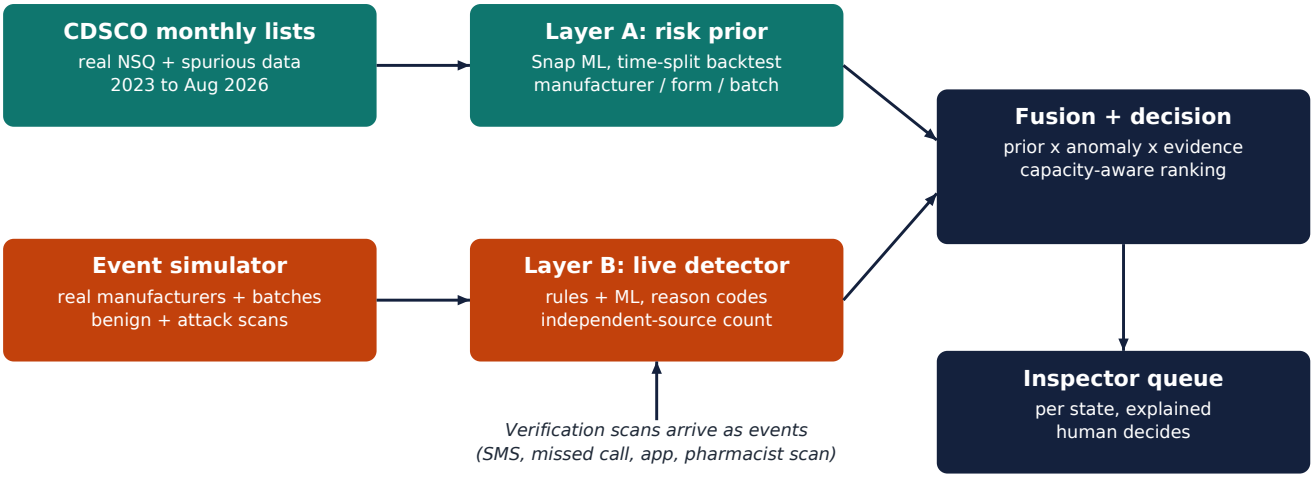
The gap. Published lists arrive after the fact and cover only tested samples. A static code can be copied onto any number of fake packs. Inspectors have no principled, evidence-based way to decide what to sample first. The people harmed are patients (especially children) and honest manufacturers whose names get attached to fakes.

2. Our solution

BatchGuard answers one operational question: **which batch should an inspector sample first this week, and why?** The output is an order, not a verdict, because testing capacity is fixed. Each layer covers the other's blind spot.

Layer alone	What it knows	Blind spot	What the other layer adds
A. Risk prior (history)	Which manufacturers, product forms and states fail most often	Nothing about what is happening now; cannot see a first-time failure on its own	A live signal that a specific batch is being cloned or behaving oddly
B. Live scan detector	How verification identities behave right now	No idea which batches are inherently risky, so every alert looks equal; can be gamed by scan flooding	Prior risk to rank alerts, plus independent evidence to resist flooding

3. Architecture



Teal = real data Amber = simulated, labelled Navy = decision output

Component	Job	Deliverable
Data ingestion	Download and parse CDSCO monthly NSQ and spurious lists (central and state) into one table	Clean table with source file and month; parse rejects counted
A. Risk prior	Score manufacturers, product forms and batches for likelihood of failing	Snap ML model, two baselines, chronological backtest
Event simulator	Generate scans with real manufacturers, batches and states; inject benign and attack behaviour	Seeded, open, labelled generator
B. Live detector	Per-identity features, rule baseline, ML detector, reason codes	Streaming service on IBM LinuxONE
Fusion + decision	Combine prior risk, live anomaly and evidence quality; rank under an inspection budget	Ranked queue per state; action ladder
Privacy + dashboard	Pseudonymised identities, coarse location, live feed and queue view	Inspector dashboard

4. Layer A: the real-data core

- **Data.** CDSCO monthly NSQ and spurious-drug lists: product, batch number, manufacturing and expiry dates, manufacturer, failure reason, reporting source and month. Fields and at least 12 usable months are confirmed in the first two hours of the build.
- **Features.** Manufacturer failure history (counts, recency), product category and dosage form (for example oral liquids against injectables), failure type, state, batch age at test, and a missing-state flag. Rates are used wherever a month publishes the number of samples tested.
- **Model and split.** Snap ML ranking model trained on lists to August 2025 and tested on September 2025 to August 2026, a chronological split with no leakage across time.
- **Baselines.** Random selection and "most past failures". A predictive claim is made only if BatchGuard beats the stronger of the two on held-out months.
- **First-time failures.** A manufacturer with no earlier failure is invisible to history alone. Layer A carries product-form risk as a coarse prior for such cases, and the Coldrif backtest reports plainly what history would and would not have shown. The live layer exists for exactly this gap.
- **Sampling bias.** Only tested batches appear, and some states submit nothing in some months. The queue shows denominators and flags missing states instead of hiding them.

5. Layer B: live clone-suspicion detection

- **Simulated, labelled stream.** Verification scans seeded with real manufacturer and batch identities. No simulated result is presented as real-world counterfeit-detection performance.
- **Benign behaviour is simulated too.** Repeat scans by the same buyer, offline sync delays, pharmacist bulk checks and legitimate travel, so an unusual scan is not automatically treated as suspicious.

- **Attack patterns.** Code copied across distant locations, bursts of scans on one identity, a slow adaptive attacker who stays under rate thresholds, and scan flooding to create false alerts. At least one attack type is held out from the detector's design and used only for testing.
- **Detector.** Per-identity features (scan velocity, geographic spread, source diversity, time pattern) feed a rule baseline and an ML detector. Every alert carries a reason code.
- **Flood resistance.** Rate limits, independent-source counts and required corroboration: many scans from one source count for little, a few scans from many independent sources count for more.

6. Fusion and the human decision

Each queue item shows three separate values, not one opaque score: **historical risk**, **scan anomaly** and **evidence quality** (how many independent sources, how recent, how consistent). The combined priority is a triage aid and always lists its reason codes, for example *clone pattern*, *manufacturer history*, *no sample yet*.

Capacity-aware ranking. The inspector sets how many samples can be drawn this week. BatchGuard returns the best list of that size, and the dashboard shows how much expected coverage each extra sample would add.

Action	When	Who acts
Accept	Low prior risk, no anomaly, good evidence	Nobody; logged
Warn the buyer	Anomalous scan, weak corroboration	Buyer sees a caution message on verification
Add to sampling queue	Elevated combined priority	Drug inspector draws and tests a sample
Escalate with evidence	High priority with several independent sources	State drug controller, with the reason-code evidence pack

Safety rule: BatchGuard never blocks a medicine automatically and never declares a pack counterfeit from a score. Doubt sends a human to test a sample.

7. Role of IBM LinuxONE

- Verification scans run as a continuous event stream through a scoring service on LinuxONE; Snap ML is used for training and scoring.
- The same defined workload is run on LinuxONE and on a laptop. We record event count, duration, sustained events per second, p95 and p99 latency, and software versions.
- We report only what actually ran on LinuxONE. Platform security features are mentioned only if configured and tested, and no speed advantage is claimed beyond what the measurements show.

8. Evaluation plan and success measures

Area	Measure	Compared against
Layer A ranking	Failures captured per 100 samples; Precision@K and Recall@K on held-out Sep 2025 to Aug 2026	Random; "most past failures"
Layer A backtest	Coldrif case: what history alone would have ranked, stated honestly	n/a
Layer B detection	Precision, recall, PR-AUC; false alarms per 1,000 genuine scans; detection latency	Rule-only baseline; held-out attack type; slow adaptive attacker
Fused queue	Failures plus simulated clones captured per 100 samples under a fixed budget	Layer A alone; Layer B alone
Streaming	1,000+ generated events end to end; events per second; p95 and p99 latency	Laptop run of the same workload
Explainability	Every high-priority item shows reason codes and the three separate signal values	n/a
Reproducibility	Documented commands rerun the evaluation and reproduce reported numbers	n/a

9. Safety, privacy and honest limits

- **Privacy.** Scan identities are pseudonymised and location is coarsened to district level. No personal profile of buyers is built.
- **Triage, not proof.** Scores are triage estimates, not proof of safety, quality failure or counterfeiting. The dashboard states this.
- **Sampling bias.** Published lists reflect tested samples, not all batches. Coverage gaps are shown, not hidden.
- **Simulated scan layer.** It tests system behaviour under stated conditions and does not establish real-world detection performance.
- **Benign anomalies.** Repeat or distant scans can be legitimate; corroboration, rate limits and human review handle them.

10. Impact and path to a pilot

- **Who acts.** State drug controllers and CDSCO inspectors, through a ranked sampling queue, on Monday morning.
- **Who benefits.** Patients, especially children who take liquid medicines; honest manufacturers whose names are attached to fakes; and inspectors with limited capacity.
- **Pilot path.** Run the risk layer on published monthly lists for one state, with inspectors comparing the queue against their usual sampling plan. The scan layer would plug into real verification traffic only with regulator and manufacturer agreement.

11. 24-hour delivery plan

Hours	Deliverable	Acceptance check
0 to 2	Data and platform check	Lists parse and labels are understood; minimal service runs on LinuxONE
2 to 5	Historical baseline	Chronological split and baseline metrics reproduce
5 to 9	Risk model and scan detector	One model, one detector, reason codes, top-K comparison
9 to 14	API, event stream and queue UI	A new scan event reaches scoring and appears in the queue end to end
14 to 18	Deploy and benchmark	1,000 events processed; latency and throughput recorded
18 to 21	Test and evidence capture	Metrics rerun cleanly; normal and attack scenarios work; failure cases documented
21 to 24	Freeze and present	Working demo, evaluation chart, architecture, limitations, backup recording

Scope control. The smallest complete vertical slice is built first. If batch-level labels prove unreliable, the ranking narrows to manufacturer and product-form level rather than overstating batch-level prediction. Optional views are cut before the core evaluation.

12. Demo walkthrough

- Open on the Coldrif timeline: what regulators knew, and when.
- Show the Layer A chart: failures captured per 100 samples against random and "most past failures".
- Start the live stream. A benign repeat scan passes quietly; a cloned identity appears from two distant locations and rises in the queue with its reason codes.
- Move the inspector-capacity slider and watch the queue and expected coverage change.
- Close on one honesty slide: what is real, what is simulated, what ran on LinuxONE, and the measured events per second and p99 latency.